

Computer Science (J277)

J277/02, Computational thinking, algorithms and programming, June 2022

Mr. Crean

Please note that you may see slight differences between this paper and the original.

Candidates answer on the Question paper.

OCR supplied materials:

Additional resources may be supplied with this paper.

Other materials required:

- Pencil
- Ruler (cm/mm)

Duration: 90 mins

INSTRUCTIONS TO CANDIDATES

- Write your name, centre number and candidate number in the boxes above. Please write clearly and in capital letters.
- Use black ink. HB pencil may be used for graphs and diagrams only.
- Answer **all** the questions, unless your teacher tells you otherwise.
- Read each question carefully. Make sure you know what you have to do before starting your answer.
- Where space is provided below the question, please write your answer there.
- You may use additional paper, or a specific Answer sheet if one is provided, but you must clearly show your candidate number, centre number and question number(s).

INFORMATION FOR CANDIDATES

- The quality of written communication is assessed in questions marked with either a pencil or an asterisk. In History and Geography a *Quality of extended response* question is marked with an asterisk, while a pencil is used for questions in which *Spelling, punctuation and grammar and the use of specialist terminology* is assessed.
- The number of marks is given in brackets [] at the end of each question or part question.
- The total number of marks for this paper is **80**.
- The total number of marks may take into account some 'either/or' question choices.

- 1(a) Tick (✓) **one** box in each row to identify whether the OCR Reference Language code given is an example of selection or iteration.

OCR Reference Language code	Selection	Iteration
for i = 1 to 10 print(i) next i		
whilescore != 0 playgame() endwhile		
if playerHit() then score = 0 endif		
switch bonus: case 0: score = 9 case 1: score = 7 case 2: score = 5 endswitch		

[4]

- (b) Write pseudocode to increment a value held in a variable score by one.

[1]

- (c) State the name of each of the following computational thinking techniques.

Breaking a complex problem down into smaller problems.

Hiding or removing irrelevant details from a problem to reduce the complexity.

[2]

2(a) A fast food restaurant offers half-price meals if the customer is a student or has a discount card. The offer is not valid on Saturdays.

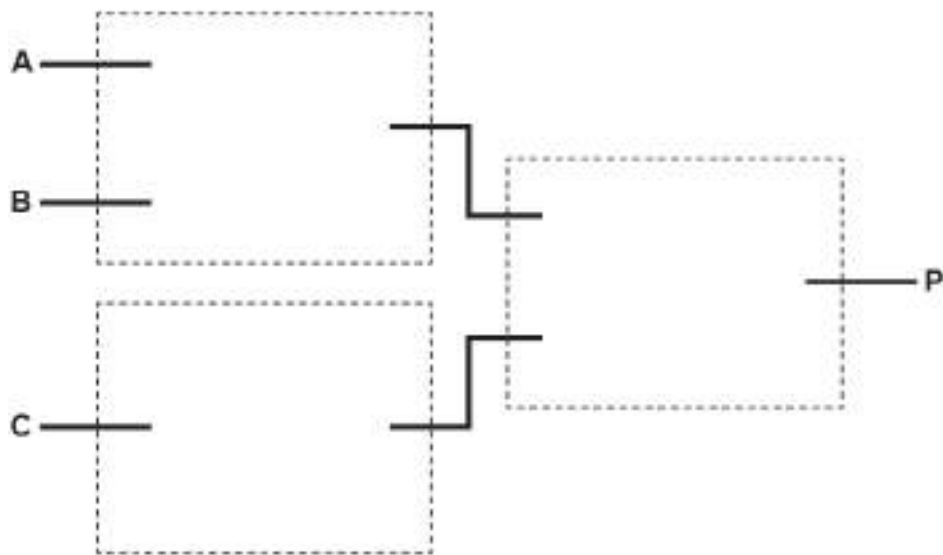
A computer system is used to identify whether the customer can have a half-price meal.

The table identifies the three inputs to the computer system:

Input	Value
A	Is a student
B	Has a discount card
C	The current day is Saturday

The logic system $P = (A \text{ OR } B) \text{ AND NOT } C$ is used.

i. Complete the following logic diagram for $P = (A \text{ OR } B) \text{ AND NOT } C$ by drawing one logic gate in each box.



[3]

ii. A truth table can be produced for this logic circuit.

Describe the purpose of a truth table.

- iii. State how many rows (excluding any headings) would be required in a truth table for the logic expression:

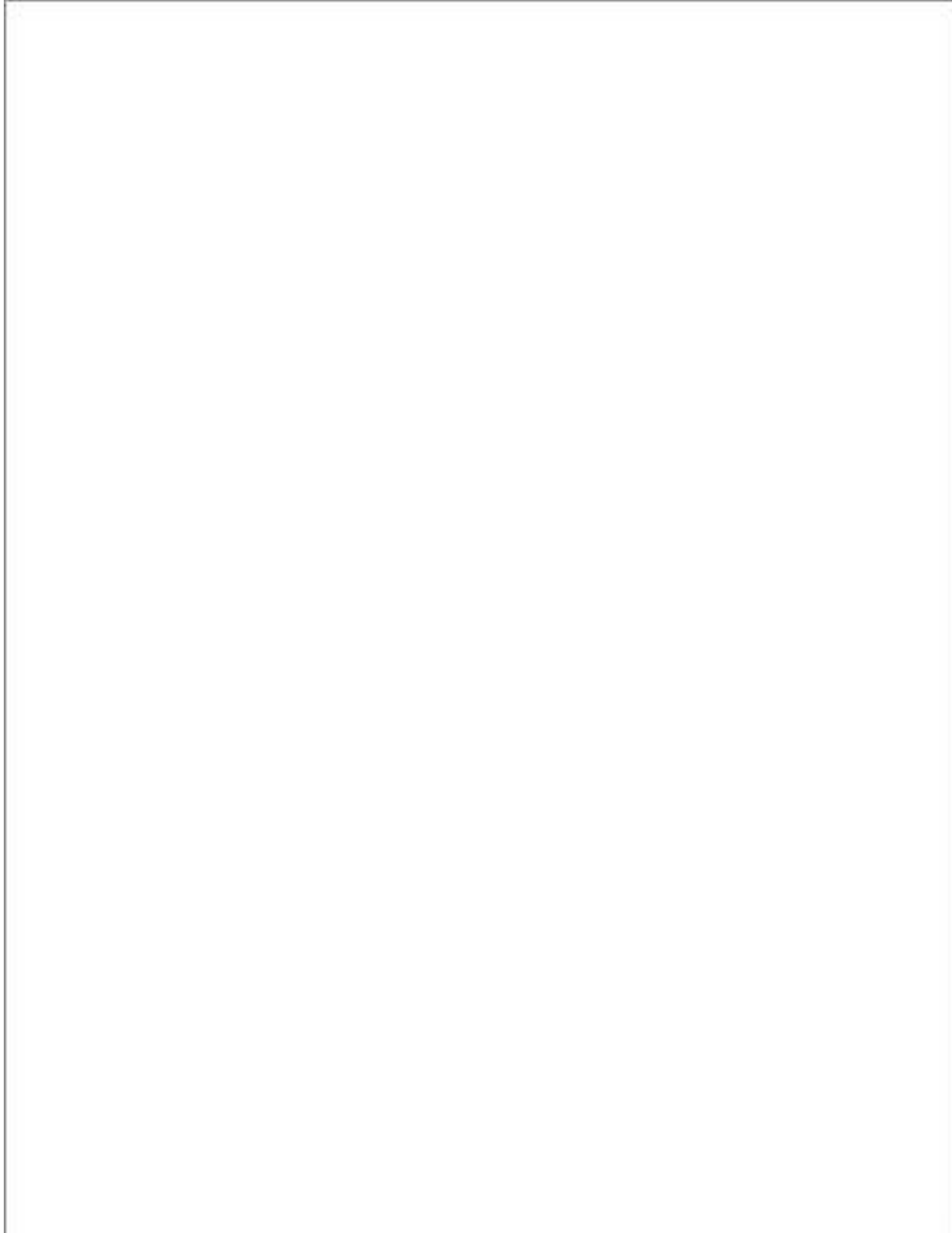
$$P = (A \text{ OR } B) \text{ AND NOT } C$$

[1]

(b) The restaurant needs an algorithm designing to help employees work out if a customer can have a half price meal or not. It should:

- input required data
- decide if the customer is entitled to a discount
- output the result of the calculation.

Design the algorithm using a flowchart.



[5]

- (c) The restaurant adds a service charge to the cost of a meal depending on the number of people at a table. If there are more than five people 5% is added to the total cost of each meal.

Customers can also choose to leave a tip, this is optional and the customer can choose between a percentage of the cost, or a set amount.

Identify **all** the additional inputs that will be required for this change to the algorithm.

[2]

- (d) Each member of staff that works in a restaurant is given a Staff ID. This is calculated using the following algorithm.

```
01 surname = input("Enter surname")

02 year = input("Enter starting year")

03 staffID = surname + str(year)

04 while staffID.length < 10

05 ? staffID = staffID + "x"

06 endwhile

07 print("ID " + staffID)
```

- i. Define the term **casting** and give the line number where casting has been used in the algorithm.

Definition -----

Line number -----

[2]

- ii. Complete the following trace table for the given algorithm when the surname “Kofi” and the year 2021 are entered.

You may not need to use all rows in the table.

Line number	surname	year	staffID	Output
01	Kofi			
02		2021		

[4]

- 3(a) A program stores the following list of positive and negative numbers. The numbers need sorting into ascending order using a merge sort.

45	12	-99	100	-13	0	17	-27
----	----	-----	-----	-----	---	----	-----

The first step is to divide the list into individual lists of one number each. This has been done for you.

Complete the merge sort of the data by showing each step of the process.

45	12	-99	100	-13	0	17	-27
----	----	-----	-----	-----	---	----	-----

[3]

- (b) Once a list of numbers are in order, a binary search can be run on the data.

Describe the steps a binary search will follow to look for a number in a sorted list.

[illegible]

- (c) A linear search could be used instead of a binary search.

Describe the steps a linear search would follow when searching for a number that is **not** in the given list.

[2]

4(a) Jack is writing a program to add up some numbers. His first attempt at the program is shown.

```
a = input("Enter a number")  
  
b = input("Enter a number")  
  
c = input("Enter a number")  
  
d = input("Enter a number")  
  
e = input("Enter a number")  
  
f = (a + b + c + d + e)  
  
print(f)
```

Give **two** ways that the maintainability of this program could be improved.

[2]

(b) Jack’s program uses the addition (+) arithmetic operator. This adds together two numbers.

i. State the purpose of each of the arithmetic operators in the table.

Arithmetic operator	Purpose
*	
/	

[2]

ii. Complete the description of programming languages and translators by writing the correct term from the box in each space.

continues	crashes	debugging	error	executable
high-level	interpreter	language	low-level	many
no	one	stops	with	without

Jack writes his program in a language. This needs to be translated into assembly or machine code before it can be executed. This is done using a translator.

One type of translator is an interpreter. This converts one line of code and then executes it, before moving to the next line. It when an error is found, and when corrected continues running from the same position. This translator is helpful when debugging code.

A second type of translator is a compiler. This converts all of the code and produces an error report. The code will not run until there are errors.

The file produced can be run the compiler.

[5]

(c) Jack decides to improve his program. He wants to be able to input how many numbers to add together each time the algorithm runs, and also wants it to calculate and display the average of these numbers.

Write an algorithm to:

- ask the user to input the quantity of numbers they want to enter and read this value as input
- repeatedly take a number as input, until the quantity of numbers the user input has been entered
- calculate and output the total of these numbers
- calculate and output the average of these numbers.

----- [6]

5(a) Customers at a hotel can stay between 1 and 5 (inclusive) nights and can choose between a basic room or a premium room.

A typical booking record is shown in the table:

firstName	Amaya
surname	Taylor-Ling
nights	3
room	Premium
stayComplete	False

i. State the most appropriate data type for the following fields:

Nights -----
Room -----

[2]

ii. Give the name of **one** field that could be stored as a Boolean data type.

----- [1]

iii. Booking records are stored in a database table called TblBookings .

The following SQL statement is written to display all customer bookings that stay more than one night.

```
SELECT ALL  
  
FROM TblBookings  
  
IF Nights < 1
```

The SQL statement is incorrect.

Rewrite the SQL statement so that it is correct.

[4]

(b) When a new booking is recorded, the details are entered into a program to validate the values. The following criteria are checked:

- firstName and surname are not empty
- room is either “basic” or “premium”
- nights is between 1 and 5 (inclusive).

If any invalid data is found “NOT ALLOWED” is displayed.

If all data is valid “ALLOWED” is displayed.

i. Complete the following program to validate the inputs.

You must use **either**:

- OCR Exam Reference Language, or
- a high-level programming language that you have studied.

```
firstName = input("Enter a first name")
```

```
surname = input("Enter a surname")
```

```
room = input("Enter basic or premium")
```

```
nights = input("Enter between 1 and 5 nights")
```

```
stayComplete = False
```

Blank lined paper for writing.

ii. Complete the following test plan to check whether the number of nights is validated correctly.

Test data (number of nights)	Type of test	Expected output
2		ALLOWED
	Boundary	ALLOWED
	Erroneous / Invalid	NOT ALLOWED

[3]

(c) A Basic room costs £60 each night. A Premium room costs £80 each night.

- i. Create a function, `newPrice()`, that takes the number of nights and the type of room as parameters, calculates and returns the price to pay.

You do **not** have to validate these parameters.

You must use **either**:

- This image shows a blank sheet of white paper with horizontal dashed lines, similar to notebook paper. The lines are evenly spaced and run across the width of the page. In the bottom right corner, there is a small black box containing the number [4].

You must use **either**:

- [3]

- (d) A hotel has nine rooms that are numbered from room 0 to room 8.

The number of people currently staying in each room is stored in an array with the identifier `r room`.

The index of `room` represents the room number.

Array room									
Index	0	1	2	3	4	5	6	7	8
Data	2	1	3	2	1	0	0	4	1

The following program counts how many people are currently staying in the hotel.

```
for count = 1 to 8
```

```
total = 0
```

```
total = total + room[count]
```

next count

```
print(total)
```

When tested, the program is found to contain **two** logic errors.

Describe how the program can be refined to remove these logic errors.

[2]

- (e) A hotel car park charges £4 per hour. If the car is electric, this price is halved to £2 per hour.

Write an algorithm to:

- take as input the number of hours the user has parked and whether their car is electric or not
- calculate and output the total price
- repeat continually until the user enters 0 hours.

You must use **either**:

- OCR Exam Reference Language, or
- a high level programming language that you have studied.

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

[6]

END OF QUESTION PAPER

Mark Scheme

Question		Answer/Indicative content	Marks	Guidance															
1	a	1 mark per correct row <table><thead><tr><th>OCR Reference Language code</th><th>Selection</th><th>Iteration</th></tr></thead><tbody><tr><td>for i = 1 to 10 print(i) next i</td><td></td><td>✓</td></tr><tr><td>while score != 0 playgame() endwhile</td><td></td><td>✓</td></tr><tr><td>if playerHit() then score = 0 endif</td><td>✓</td><td></td></tr><tr><td>switch bonus: case 0: score = 9 case 1: score = 7 case 2: score = 5 endswitch</td><td>✓</td><td></td></tr></tbody></table>	OCR Reference Language code	Selection	Iteration	for i = 1 to 10 print(i) next i		✓	while score != 0 playgame() endwhile		✓	if playerHit() then score = 0 endif	✓		switch bonus: case 0: score = 9 case 1: score = 7 case 2: score = 5 endswitch	✓		4 (AO2 1b)	No mark given if both boxes in a row ticked. Accept any response (ticks, crosses, etc) that clearly indicates candidate's choice. <u>Examiner's Comments</u> Candidates appeared to struggle with this question. In particular the use of switch/case was not well understood. This may be because some high-level languages such as Python have not traditionally supported this. The recent update to Python 3.10 introduces this construct. Centres who use Python may want to consider upgrading their installation to allow practical implementation of the switch/case construct.
OCR Reference Language code	Selection	Iteration																	
for i = 1 to 10 print(i) next i		✓																	
while score != 0 playgame() endwhile		✓																	
if playerHit() then score = 0 endif	✓																		
switch bonus: case 0: score = 9 case 1: score = 7 case 2: score = 5 endswitch	✓																		

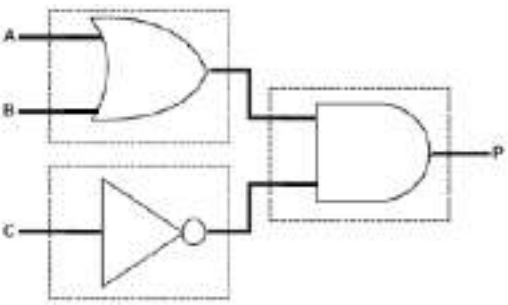
Mark Scheme

Question	Answer/Indicative content	Marks	Guidance
b	• <code>score = score + 1 // score +=1 // score++</code>	1 (AO3 2b)	Allow other logically correct answers that result in score increasing by one and being overwritten. <u>Do not accept <code>score + 1 / score = +1</code></u> Accept valid structured English answers that refer to score increasing and overwriting the existing value by one. e.g. “score becomes/equals score plus one” Ignore any superfluous code that does not affect the outcome <u>Examiner’s Comments</u> Most candidates were able to temporarily add one to the value of score (by using code such as <code>score + 1</code>). Fewer were able to assign this new value to the variable of score and therefore fully answer the question. Answers such as <code>score = score + 1</code>, <code>score++</code> or <code>score += 1</code> were all accepted, as were any longer responses that used intermediate variables. As long as the value of score ended up being incremented by one, examiners were instructed to give credit.  Assessment for learning One notable response that was not allowed was <code>score =+1</code>. This statement assigns the value of ‘positive 1’ to score, overwriting the previous value held. This is a good example of the precision required to gain marks. Candidates with practical programming experience are more likely to recognise this.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
	c		• Decomposition • Abstraction	2 (AO1 1a)	Correct answer only. Ignore spelling errors. <u>Examiner's Comments</u> This question was answered well by the majority of candidates.
			Total	7	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
2	a	i	• A OR B • NOT C • AND gate 	3 (AO2 1b)	1 mark per gate. Correct symbols must be used. NOT gate must have circle for inversion, OR and AND must not have a circle Mark the shape of each gate, not the name written if given. Ignore any writing / notes. Lines do not have to be drawn or joined up, but if they are, gates must have the correct number of inputs/outputs. Penalise once then FT. <u>Examiner's Comments</u> This question was answered extremely well. Candidates were familiar with the logic symbols required for each gate. A few candidates missed the circle from the NOT gate symbol (which turned this into a buffer instead) or added a circle to the AND and OR gates. This turned them into NAND and NOR gates. These logic gates are not on the GCSE specification, but are covered at A Level. Teachers may find it beneficial to briefly discuss these in order to anticipate mistakes like these. A small number of candidates drew indistinct logic gate symbols and named them instead, such as a rectangle with AND written in; this was not credited with a mark.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
		ii	- To show all possible inputs (to the logic circuit)... ...and the associated/dependent output (for each input)	2 (AO1 1b)	For 2nd BP, must be clear that the output is linked to the input values given. “All possible combinations of inputs and outputs” gains the first mark (all possible inputs) but not the second. “The output for each possible input” gains both marks <u>Examiner’s Comments</u> It is clear that candidates have experienced and used truth tables with logic circuits. However, the majority were not able to describe the truth table’s purpose with precision. A truth table defines the expected outputs for a logic circuit depending on the inputs given. Furthermore, the truth table covers all possible permutations of inputs. A logic circuit with three inputs (A,B and C) will have 8 possible rows in a truth table to cover the 8 possible ways that True and False values for three inputs can be arranged. Misconception The number of rows in a truth table depends on the number of inputs. The number of rows can be given by 2^x, where x is the number of inputs. Therefore, a truth table with 3 inputs would have 2^3 rows, or $2 \times 2 \times 2 = 8$ rows. Exemplar 1 Here the candidate has achieved both marks available; it is clear that every variation of input is included in the truth table and the output is very clearly linked to these inputs.
		iii	8 // eight	1 (AO2 1a)	Accept other answers that equate to 8 (e.g. 2^3)

Mark Scheme

Question		Answer/Indicative content	Marks	Guidance																																													
	b	- Start and end/stop with all boxes connected, no boxes that do not lead to another box (no arrows needed) - Input three variables using parallelogram shape - Checks all three criteria (day, student, discount card) using diamond shape(s) with two lines from each ...Outputs "full price" <u>with correct conditions</u> using parallelogram shape ...Outputs "half price" <u>with correct conditions</u> using parallelogram shape <u>Guidance for correct outputs</u> <table><tr><th colspan="2">Conditions</th><th>Outcome</th></tr><tr><td colspan="2">Not Saturday and (either a student or has a discount card).</td><td>Half price</td></tr><tr><td colspan="2">Saturday or (not a student and doesn't have a discount card).</td><td>Full price</td></tr></table> <table><tr><th>Saturday</th><th>Student</th><th>Discount Card</th><th>Outcome</th></tr><tr><td>N</td><td>N</td><td>N</td><td>Full price</td></tr><tr><td>N</td><td>N</td><td>Y</td><td>Half price</td></tr><tr><td>N</td><td>Y</td><td>N</td><td>Half price</td></tr><tr><td>N</td><td>Y</td><td>Y</td><td>Half price</td></tr><tr><td>Y</td><td>N</td><td>N</td><td>Full price</td></tr><tr><td>Y</td><td>N</td><td>Y</td><td>Full price</td></tr><tr><td>Y</td><td>Y</td><td>N</td><td>Full price</td></tr><tr><td>Y</td><td>Y</td><td>Y</td><td>Full price</td></tr></table>	Conditions		Outcome	Not Saturday and (either a student or has a discount card).		Half price	Saturday or (not a student and doesn't have a discount card).		Full price	Saturday	Student	Discount Card	Outcome	N	N	N	Full price	N	N	Y	Half price	N	Y	N	Half price	N	Y	Y	Half price	Y	N	N	Full price	Y	N	Y	Full price	Y	Y	N	Full price	Y	Y	Y	Full price	5 (AO3 2a)	<pre>graph TD; Start([Start]) --> InputStudent[/INPUT student/]; InputStudent --> InputDiscCard[/INPUT discCard/]; InputDiscCard --> InputDay[/INPUT day/]; InputDay --> Decision1{IF day NOT "Saturday"}; Decision1 -- TRUE --> Decision2{IF student or discCard}; Decision1 -- FALSE --> OutputFull[/OUTPUT "full price"/]; Decision2 -- TRUE --> OutputHalf[/OUTPUT "half price"/]; Decision2 -- FALSE --> OutputFull; OutputHalf --> End([End]); OutputFull --> End;</pre> Question asks for a flowchart. Answers as pseudocode, high level language or other forms are not acceptable 9 (NAQ). BP 4 and 5 only to be awarded if all decisions ensure correct output and <u>clear what the decisions are</u>. FT for incorrect shapes used or no inputs as long as decisions are logically correct. Must attempt all three decisions. Allow calculation of half price / full price instead of message but this must still be output. Inputs / decisions may be presented as individual or combined boxes but must still store as three variables. Penalise lack of parallelogram for input/output once only then FT BOD parallelogram shapes if not sure whether input or output as long as context is clear (e.g inputs at start, outputs at end) <u>Examiner's Comments</u> Most candidates who answered this question were comfortable using the correct flowchart symbols listed in the specification. A mark was available for
Conditions		Outcome																																															
Not Saturday and (either a student or has a discount card).		Half price																																															
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Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
					including suitable start / end symbols and connecting all other symbols, so even candidates who may struggle should feel confident of being able to access some marks. Marks were dropped when responses did not include suitable inputs to the algorithm. The first bullet point in the question stem was clear that these were required. Candidates achieving lower scores tended to group decisions so that any processing was removed (such as “can they have a discount?”). More successful responses decomposed the problem and used a succession of smaller decisions (“do they have a discount card?”, “are they a student?”, “is it Saturday?”). These decisions then point towards the correct outputs.
	c		- Number of people (at the table) // whether there are more than 5 people or not - Choice between percentage and value // actual value of both percentage, value	2 (AO32a)	Ignore additional inputs that would be sensible, such as cost of the meal. Accept inputs in form of pseudocode / high-level language. Max 1 if other irrelevant inputs given. "Whether to leave a tip or not" or "Amount of tip" NE for BP2. Must address both the percentage and value of tip if asked for. BOD "type of tip" for BP2 <u>Examiner's Comments</u> This question type is new to the J277 specification and asked candidates to list inputs that will be required as part of the planning stage for an algorithm. Successful candidates were able to identify the raw data needed from the user in order to be able to solve the problem. Unsuccessful responses tended to simply rewrite the question or miss out key information.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance																																			
	d	i	- Convert/change one data type to another - Line 03 // 3 // three	2 (AO1 1b) (AO2 2b)	Do not accept "change to string" - this is the use in this example but not a definition. <u>Examiner's Comments</u> Many candidates correctly defined casting as changing data from one data type to another. Some candidates defined this term as changing a variable from an integer to a string, which is only one example of what can be done and not a definition. The majority of candidates then gave the correct line number (line 03) for there this was shown the example code given.																																			
		ii	- Kofi2021 as staffID on line 03 - Kofi2021x as staffID on line 05 - Kofi2021xx as staffID on line 05 - ID Kofi2021xx output on line 07 as first and only output	4 (AO3 2c)	Max 2 if incorrect order. Ignore misspelling of Kofi Penalise lack of / errors with line numbers once then FT. Ignore capitalisation. Ignore additional lines unless outcome impacted. staffID does not have space in. Output does have a space in. Penalise spaces once then FT. Do not penalise unless obvious. Quotes around answer is OK, but do not allow quotes around partial answers, e.g. "ID" Kofi2021xx is incorrect. <table><tr><th>Line number</th><th>sur name</th><th>year</th><th>staffID</th><th>Output</th></tr><tr><td>01</td><td>Kofi</td><td></td><td></td><td></td></tr><tr><td>02</td><td></td><td>2021</td><td></td><td></td></tr><tr><td>03</td><td></td><td></td><td>Kofi2021</td><td></td></tr><tr><td>05</td><td></td><td></td><td>Kofi2021x</td><td></td></tr><tr><td>05</td><td></td><td></td><td>Kofi2021xx</td><td></td></tr><tr><td>07</td><td></td><td></td><td></td><td>ID Kofi2021xx</td></tr></table> <u>Examiner's Comments</u> This question asked candidates to trace through a given algorithm to show the value of three variables at various points in the algorithm.	Line number	sur name	year	staffID	Output	01	Kofi				02		2021			03			Kofi2021		05			Kofi2021x		05			Kofi2021xx		07				ID Kofi2021xx
Line number	sur name	year	staffID	Output																																				
01	Kofi																																							
02		2021																																						
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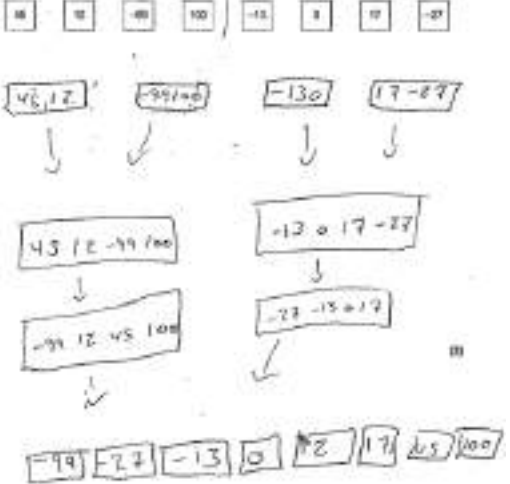
Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
					The algorithm itself was relatively simple. It used condition-controlled iteration to repeat while the length of the username was less than 10 characters. Most candidates gained the first 2 marks for the initial changes to <code>staffID</code>. However few candidates were able to trace through the iteration and conclude that the final username should end up as <code>ID Kofi2021xx</code>. Marking this question considered the spaces within the username at various points. The algorithm results in one space only, in between <code>ID</code> and <code>Kofi2021xx</code>. Where extra spaces appeared or were missed, this was penalised. However, examiners were instructed to give clear benefit of doubt, and to only do this if the space was clearly present/missing. It is important to understand that “ab” and “a b” are two strings that are not the same. This level of precision should be encouraged within GCSE Computer Science. Experience of practical programming will help reinforce the impact of spaces within programming and algorithms.
			Total	19	

Mark Scheme

Question		Answer/Indicative content	Marks	Guidance
3	a	• Merge into correct sorted lists of size 2 (12 45 / -99 100 / -13 0 / -27 17) • Merge into correct sorted lists of size 4 (-99 12 45 100 / -27 -13 0 17)... • ...Merge into correct sorted list of size 8 (-99 -27 -13 0 12 17 45 100)	3 (AO2 1b)	Do not credit BP3 simply for a sorted list. Groups of numbers must clearly be the correct size. Do not all allow answers that show lists being merged and then sorting in place, this is incorrect. <u>Examiner's Comments</u> Most candidates were able to correctly demonstrate the required steps of the merge sort algorithm. This involves merging the set of 8 individual lists into 4 lists of size 2, then 2 lists of size 4, then 1 list of size 8. At each point, each list produced must be in order. It is this process of merging lists together that sorts the values. No mark was given simply for showing the final list sorted unless the previous step(s) had been followed; any candidates an alternative sorting algorithm (for example a bubble sort) would achieve 0 marks even though the final outcome is the same.  Misconception Many candidates showed merged lists which were out of order to start with, and then sorted these merged lists. This is highly inefficient and not what the standard merge sort algorithm describes. In merge sort, the process of merging lists together produces the sorted list. There does not need be any second step to sort values as they should already be in order. Exemplar 2

Mark Scheme

Question	Answer/Indicative content	Marks	Guidance
			 This exemplar shows the misconception detailed previously. Here, the candidate firstly merges together the individual lists into 4 lists of size 2, but these lists are not all in order (e.g. 17, -27 is incorrect). The second step then merges these lists together into 2 lists of size 4, but again, these are not in order. An unnecessary next step is then shown where the lists are then sorted in place.
b	Any four bullet points for 1 mark each • Select / choose / pick middle number (or left/right of middle as even number) and .. • ...check if selected number is equal to / matches target number (<i>not just compare</i>) • ...if searched number is larger, discard left half // if searched number is smaller, discard right half • Repeat until number found • ... or remaining list is of size 1 / 0 (number not found)	4 (AO1 1b)	Do not allow “split the list in half” on its own as first step, this is incorrect. Can get BP1 and 2 in one step (e.g. “check if the middle number is the one we’re looking for”) For BP3, accept focussing on correct half Repeat (BP4) must be in the context of an attempt at a binary search. Allow correct reference to recursion. “until number is not in the list” is NE for final BP. Need to explain how this is known. Examiner’s Comments This question asks candidates to describe how a binary search would look for a number in a sorted list. Most candidates answered this well. The most common mark for this question was 3 out of 4. The majority of candidates clearly understood that the middle value is chosen and compared against the value being searched for. However, almost all

Mark Scheme

Question	Answer/Indicative content	Marks	Guidance
			candidates then went on to discuss which half of the list should be discarded or focused on based on this comparison. One possibility is that the middle value is the number we are searching for. Few candidates discussed this possibility. As the process repeats, this check is important if the number is ever to be found! Assessment for learning When discussing a binary search there are three possibilities when comparing the middle value to the number being searched for.: 1 the middle value is higher than needed (in which case the left half of the list should be focused on) 2. the middle value is lower than needed (in which case the right half of the list should be focused on) 3. the middle value is equal to the value needed (in which case the value has been found and the process can stop. Most responses only considered the first two of these possibilities. This is worth discussing when teaching this topic.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
	c		1 mark each - Starting with the first value - checking all values in order	2 (AO1 1b)	2nd bullet point must cover both ideas of checking all of the values AND being done in order. “Checks each value” / “one by one” / “step by step” by itself is NE, does not say in order. Do not accept “repeat until value found” for BP2 (question says number is not in the list) “Checks each value from beginning to end” implies order so gets both BP1 and BP2. <u>Examiner’s Comments</u> Responses to this question generally lacked precision. The question specifically asks about searching for a number that is not in the list. However, many candidates discussed stopping when the value had been found. This is clearly not possible given the scenario. Many responses discussed checking each number but did not state whether this would be in a particular order. Some candidates discussed checking values randomly, or even from the right down to the left. For this questions, candidates may have known that a linear search starts at the left most number and then checks each value in the list in order until reaching the end of the list. However, few responses stated this or anything close to this. This level of precision (and conversely, not assuming that steps are obvious) should be encouraged within GCSE Computer Science.
			Total	9	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
4	a		Any two bullet points for one mark each: • Add comments • Name variables sensibly • Put into subroutine / procedure / function • Use loop / iteration	2 (AO2 1b)	Do not accept indentation (no code to sensibly indent in this example) “Use a subroutine” is not enough. Must be clear that existing code will be put into a new subroutine. <u>Examiner’s Comments</u> This question asked about maintainability. It was important that a candidate’s response referred to the code given. “Give two ways that maintainability of a program could be improved” is a different question than the one asked. Because of this, responses that were accepted must genuinely improve the maintainability of the program shown. One very common wrong response was indentation; although this would be useful generically, there was no code presented that would benefit from being indented. Many candidates were able to give responses such as use of comments and sensibly named variables that would genuinely improve the given program. Where candidates described putting the given code inside a newly defined subroutine. This response was credited. However some responses simply said “use a subroutine”. This was not enough.
	b	i	• Multiplication • Division	2 (AO1 1a)	Accept other correct answers that mean the same Accept floor / integer division // division with no remainder (Python v2.x) <u>Examiner’s Comments</u> This question was answered well by the majority of candidates.
		ii	• high-level • stops // crashes • no • executable • without	5 (AO1 1b) (AO2 1b)	Ignore spelling errors. <u>Examiner’s Comments</u> This question was answered well by the majority of candidates.

Mark Scheme

Question	Answer/Indicative content	Marks	Guidance
c	- input <u>and stores/uses</u> value <u>with message</u> - attempt at repeating... ... <u>correctly</u> repeats number of times given as input ... <u>correctly</u> take number as input within loop <u>and</u> calculates total of these numbers ... <u>correctly</u> calculate an average (total/num) - Output <u>both</u> total and average	6 (AO3 2b) (AO3 2c)	e.g. <pre>num = input("Enter how many numbers") for x = 1 to num temp = input("Enter a number") total = total + temp next x print(total) print(total / num)</pre> If flow chart used, correct shapes needed. Allow tolerance of 1 with number of loops for BP3 with for loops BP1 requires input with a message (can be two statements, e.g. print and then input or combined. Input must be stored or used. BP3, 4, 5 must be logically correct to be credited Ignore non-initialisation of total BP 5 can be given as FT as long as an attempt has been made at working out a total within the loop. BP6 can be given as FT long as attempt made at total <u>and</u> average (not necessarily in a loop) <u>Examiner's Comments</u> As this question appeared in Section A, candidates are free to respond in any suitable way, including using flowcharts, structured English, pseudocode or a high-level language. The majority of candidates who scored highly tended to use a high-level language consistently. The question is already decomposed for candidates and many were able to use these bullet points to build a solution that achieved the majority (or all) marks available.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
					Where mistakes were made, these tended to be with repeating code. For example, many candidates repeated the process of adding values up without repeatedly asking for a new number (and therefore continually adding the same number). Other candidates used condition-controlled iteration to repeat the process but did not make sure that the condition being evaluated ever returned a False value, therefore repeating infinitely. Where a mistake was made in one section (such as with iteration), examiners were instructed to use FT (follow through) marks where possible if later sections were logically constructed. This made sure that marks could be given where appropriate. A few candidates were not able to access many of the marks available in this question, suggesting that they would benefit from more practical programming time in lessons.
			Total	15	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
5	a	i	- Integer - String	2 (AO3 2a)	Accept other valid data types from high-level languages (e.g. byte, short for integers) Do not accept descriptions (e.g. "whole number", "text"). Do not accept "character(s)" for string.
		ii	stayComplete	1 (AO3 2a)	Ignore spaces or misspelling as long as recognisable. <u>Examiner's Comments</u> These questions tested candidates' knowledge of data types and it was clear that this knowledge was well understood. The majority of candidates were able to correctly identify suitable data types in section (i) and identify stayComplete as the field that would be stored as a Boolean data type.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
		iii	• SELECT FirstName, Surname, Nights, Room, StayComplete // SELECT* • FROM TblBookings • WHERE • Nights > 1 // Nights >= 2 // Nights BETWEEN 2 AND 5.	4 (AO3 1) (AO3 2c)	Order of fields for BP1 not important but must show all fields and be separated by commas. Ignore capitalisation and spacing. Spelling must be correct. Ignore quotes around numeric values or field/table names. Allow other logically valid SQL statements. Check with TL if required. Ignore reference to stayComplete or other valid SQL code that would not affect output. Max 3 if in wrong order or if includes any extra invalid code <u>Examiner's Comments</u> This question tested candidates' ability to refine and rewrite incorrect code given to them. It is important to note that although the SQL statement as a whole is incorrect, not all components are incorrect; in this case, the FROM clause is correct and candidates who made no change to this line were credited. SELECT ALL is invalid SQL and should have been written to instead include all fields from the table, separated by commas SELECT * was equally accepted as a suitable response. IF is not a valid SQL keyword and needs to be replaced with WHERE. The criteria for this statement was also incorrect. The comparison symbol is incorrect and should read Nights > 1. Most candidates gained some marks on this question. The most common response corrected the criteria and not modifying the FROM clause.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
	b	i	- Checks that both firstname and surname are not empty... - Checks that room is either "basic" or "premium"... - Checks that nights is between 1 and 5 (inclusive)... ...Outputs "NOT ALLOWED" (or equivalent) if <u>any</u> of the 3 checks are invalid (must check all three) ...Outputs "ALLOWED" (or equivalent) <u>only</u> if all three checks are valid (must check all three) <i>Note : output marks are given for if entire system produces the correct output. For example, If a user enters a valid name and room but an invalid number of nights, the system should say "NOT ALLOWED" (or equivalent). If this works and produces the correct response no matter which input is invalid, BP4 should be given.</i> <i>The same process holds for the valid output - if (and only if) three valid inputs results in an output saying "ALLOWED" (or equivalent), BP5 should be given. Do not give this if ALLOWED is printed when (for example) two inputs are valid and one is invalid.</i> <i>For any output marks to be given, a sensible attempt must have been made at all three checks. These may not be completely correct (and may have been penalised in BPs 1 to 3) but should be enough to allow the FT marks for output.</i>	5 (AO3 2a)	Must have some attempt at <u>all three checks</u> to give output mark(s). Check for nights must check both upper and lower limits. Iteration can be used as validation if input repeatedly asked for until valid answer given. <u>Do not accept</u> logically incorrect Boolean conditions such as <code>if firstname or surname == ""</code> Do not accept $\geq$ or $\leq$ for $\geq$, $\leq$. Ignore capitalisation e.g. <pre>valid = True if firstname == "" or surname == "" then valid = False end if if room != "basic" and room != "premium" then valid = False endif if nights < 1 or nights > 5 then valid = False endif if valid then print("ALLOWED") else print("NOT ALLOWED") endif</pre> BP1 to 3 can check for valid or invalid inputs. . Pay particular attention to use of AND / OR. Only give marks for output if these work together correctly. Example above shows checking for invalid data. Checks for valid data equally acceptable Examples shown below: <code>if firstname != "" and surname != ""</code> <code>if room == "basic" or room == "premium"</code> <code>if nights >= 1 and nights <= 5</code> <u>Examiner's Comments</u> This question stretched the understanding of even highly-achieving candidates and it

Mark Scheme

Question	Answer/Indicative content	Marks	Guidance
			was not uncommon to see low scoring responses. Misunderstanding of Boolean operators (AND and OR) within selection (IF) statements was something that affected a lot of candidate responses. As this question was in Section B, candidates needed to respond in OCR Exam Reference Language or a high-level language. Responses must be logically correct to gain the marks. As each check is two individual checks that both need to pass, responses can quickly get relatively complicated. As can be seen from the mark scheme, advice and examples were given to examiners to make sure that candidates who were able to successfully navigate this logic chain were credited.  Misconception Checking whether a room is either basic or premium can be done in multiple ways. Candidates can either check for the positive (i.e. check that it is either basic or premium) or check that for the negative (i.e. check whether it is something else). However, there are many common errors that were seen : • IF room == "basic" or "premium" is incorrect as the second part of the statement is not evaluated against anything. This was perhaps the most common mistake. • IF room == "basic" or room == "premium" is correct and checks for validity. • IF room == basic or room == premium is incorrect as the lack of string delimiters means that basic and room would be treated as variables rather than strings. • IF room != "basic" or room != "premium" is also incorrect. This checks for invalid input but because or is used, only one condition needs to be True for the whole statement to be True. This means that if basic is entered, it would be classed as invalid (as it isn't premium) and vice-

Mark Scheme

Question	Answer/Indicative content	Marks	Guidance
			versa. There is no way for any entry in this example to be classes as valid. • IF room != "basic" and room != "premium" is correct. This checks for invalid inputs but needs both conditions to be True. The same explanation follows for the other two necessary checks. Exemplar 3 <pre> if first Name == "*" OR surname == "*" then print("NOT ALLOWED") else if room != "basic" AND room != "premium" then print("NOT ALLOWED") else if night < 1 OR night > 5 then print("NOT ALLOWED") else print("ALLOWED") end if end if end if end if </pre> This exemplar shows a fully correct response. The candidate checks for invalid responses and correctly uses Boolean operators to check multiple criteria at each step. If any check returns True, "Not allowed" is printed and the program ends. Efficient use of if ... else ... means that the next check only proceeds if the previous check returns False. If all three checks return False, the final else is triggered to print "Allowed". It must be noted that this is only one way of achieving full marks. An equivalent program that checks for valid responses at each turn would also be possible. Candidates should be encouraged to use whatever structure they feel is sensible. If a response can logically be followed then it will achieve high marks.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance												
		ii	- Normal 1 or 5 (<i>not 0 or 6 as says allowed</i>) - Any numeric value except 1 to 5 // any non-numeric input (e.g. "bananas")	3 (AO3 2c)	Allow other descriptions that mean normal (e.g. valid / typical / acceptable) <table><tr><th>Test data (number of nights)</th><th>Type of test</th><th>Expected output</th></tr><tr><td>2</td><td>Normal</td><td>ALLOWED</td></tr><tr><td>1 // 5</td><td>Boundary</td><td>ALLOWED</td></tr><tr><td>e.g. 7</td><td>Erroneous/Invalid</td><td>NOT ALLOWED</td></tr></table> <u>Examiner's Comments</u> This question was answered well by the majority of candidates.	Test data (number of nights)	Type of test	Expected output	2	Normal	ALLOWED	1 // 5	Boundary	ALLOWED	e.g. 7	Erroneous/Invalid	NOT ALLOWED
Test data (number of nights)	Type of test	Expected output															
2	Normal	ALLOWED															
1 // 5	Boundary	ALLOWED															
e.g. 7	Erroneous/Invalid	NOT ALLOWED															
	c	i	- Function header for newPrice... ...taking (at least) two parameters ...correctly calculates price based on parameters (if present) <u>within function</u> .../ ... returns this calculated price	4 (AO3 2b)	BP1 must be clear that a new function is being defined. E.g. function / def keyword. Allow FT for subsequent marks if not present. Ignore any code outside attempt at function definition. Ignore additional parameters. Ignore inputs or additional code as long as these do not overwrite parameters or affect operation of function. If inputs used instead of parameters, FT for BP3. Allow use of else for second room type in BP3. Attempt at calculation needed to award BP4. Must return (not output) value. Return can be done e.g. in VB by assigning to function name (e.g. newPrice = price) e.g. <pre>function newPrice(nights, room) if room == "basic" then if room== 60 * nights then elseif room == "premium" then price = 80 * nights endif endif endif</pre>												

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
					return price endfunction <u>Examiner's Comments</u> Defining functions appeared to be a concept that candidates did not fully understand. Where a candidate did not attempt to define a function and instead simply calculated the price needed, very few (if any) marks were available. Successful responses could have been constructed from any suitable function definition keyword such as function (OCR ERL, VB, JavaScript, etc), def (Python) or others. Answers in C#, Java or other languages referring to methods were also accepted.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
		ii	• Call function newPrice... • ...with <u>("premium", 5)</u> as parameters • ... Output returned value	3 (AO3 2b)	Order of parameters not important "premium" must use string delimiters (e.g. "quotes") e.g. <pre>print(newPrice("premium", 5)) x = newPrice(5, "premium") print(x)</pre> Do not allow function definitions for BP1 Ignore capitalisation of newPrice Candidate could store returned value in a variable and then print this, or store parameters in variables before passing in - these are all acceptable Ignore any superfluous code given Do not credit answers where newPrice is overwritten prior to use. Ignore spaces. Allow function call if brackets missing (e.g. newprice instead of newprice()) <u>Examiner's Comments</u> Even if candidates were not able to create a function, this question was independent to (i) and so marks were available for simply using the function to output a value. Candidate found this question challenging. Many candidates called the function but most did not understand that the room type was a string and so required string delimiters (e.g. quotation marks) around the parameter. Where candidates defined local variables, assigned the values needed to the variables and then passed these into the function call were accepted.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
	d		- For loop changed to include 0 - total = 0 moved to before loop starts / removed	2 (AO3 2c)	Allow loop changed to 0 to 8 or 0 to 9 (Python) Do not accept moving total outside loop, NE (could be moved to after loop which would still be a logic error). Do not accept move to top of loop. Accept corrected code shown. Accept reference to count variable limits for BP1. <u>Examiner's Comments</u> This question acted as an excellent discriminator. The majority of candidates identified and fixed the problem with the array (starting at 0 rather than the 1 given). Fewer candidates were able to identify and fix the issue relating to the total being set to zero for each iteration. A number of candidates identified this second issue but were either not precise enough with their response (suggesting that the line could be moved but not stating where it should be moved to) or did not attempt to go beyond the simple identification. Candidates should be clear that this question required detail of how the program could be refined to fix problems, not just to identify problem(s).
	e		- Inputs hours AND electric (two separate inputs), storing or using these. - Checks if car is electric (IF/Select statement)... ... correctly calculates and outputs price (hours * 2 // price / 2) for electric ... correctly calculates and outputs price (hours * 4 // electric price * 2) for non-electric - Attempt at repetition of BP1 to 4... ...until 0 hours entered	6 (AO3 2c)	Initialisation of price and hours not necessary, but if present hours must be non-zero for BP6 to be given. BP5 must include all points attempted. Can still be credited if any of BP1 to 4 not attempted / incorrect. BP6 can be given as FT even if BP5 (loop) is in the wrong place / does not include all required code BP6 could be achieved as repeated function calls / recursion Initial input outside of loop that is then <u>also</u> included within loop is fine. For example, input of hours outside of loop but input is then repeated again at end of loop.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
					Do not accept while hours > 0 (could be -1) Do not penalise answers where 0 is output when loop exits e.g. <pre>while hours != 0 hours = input("Enter hours") electric = input("enter Y for electric or N") if electric == "Y" then price = hours * 2 elseif electric == "N" then price = hours * 4 endif print(price) endwhile</pre> <u>Examiner's Comments</u> This question was relatively well answered by candidates. Candidates were generally able to create suitable high-level program code to calculate and output the total price based on the information given. Many candidates ignored the requirement to repeat until 0 was entered; in this case, 4 out of the 6 marks were still available. To achieve marks for iteration it needed to both repeat the correct parts of the program and correctly terminate as per the requirements given. A typical mistake was to repeatedly calculate the price but not ask afresh for new inputs.
			Total	30	